NOT MEASUREMENT SENSITIVE

MIL-T-62319C
28 July 1989
SUPERSEDING
MIL-T-62319B
2 November 1984

MILITARY SPECIFICATION

TRUCKS, CARGO: DIESEL OR GASOLINE ENGINE DRIVEN, 19,000 TO 24,000 POUNDS GVW, 4X2 AND 4X4, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers diesel or gasoline engine driven, four-wheel, two-rear-wheel drive, and four-wheel, four-wheel drive cargo trucks, having minimum gross vehicle weights (GVW) of 19,000 to 24,000 pounds, for normal operating conditions. The vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

1.2 Vehicle classification. The vehicle shall be one of the classes and models shown in table I, as specified (see 6.2):

TABLE I. Vehicle classification.

*

Class	Model	Nomenclature
C D E	4x2	4 wheel, 2 wheel drive
– – E	4x4	4 wheel, 4 wheel drive

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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1.2.1 Class. The class of the vehicle shall be determined by the minimum GVW rating as follows:

	GVW
Class C	19,000 pounds
Class D	21,000 pounds
Class E	24,000 pounds

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	VV-F-800	- Fuel Oil, Diesel.
	ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).
*	CCC-C-419	- Cloth, Duck, Cotton, Unbleached Plied Yarns, Army and Numbered.

MILITARY

*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
*	MIL-M-43719	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specification For.
	MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

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MILITARY

MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MS51118	- Pintle Assembly, Towing – 40,000 Lbs. Capacity, Manual Release.
MS51335	- Pintle Assembly, Towing – 18,000 Lbs. Capacity, Manual Release.
MS75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 28-Volt, Waterproof.
MS75021	- Connector, Receptacle, Electrical -12 Contact, Intervehicular, 28-Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF DEFENSE (DoD)

*	DODISS	- Department of Defense Index of Specifications and Standards.

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DOT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

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* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Standards.

A120	- Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses (DOD adopted).

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)

Standards Manual

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, Inc.

SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DOD adopted).
J350	- Spark Arrester Test Procedure for Medium Size Engines (DOD adopted).
J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J682	- Rear Wheel Splash and Stone Throw Protection (DOD adopted).
J683	- Tire Chain Clearance - Trucks, Buses, and Combinations of Vehicles.
J688	- Truck Ability Prediction Procedure (DOD adopted).
J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs.

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J706	- Rating of Winches.
J844	- Nonmetallic Air Brake System Tubing (DOD adopted).
J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.14, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.14, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model component shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

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3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2).

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in manual transmissions, transfer cases and rear axles shall be of the permanent magnet type.

* 3.1.1.4 Wood treatment. As specified herein (see 3.5.4.1), wood surfaces shall be treated in accordance with MIL-STD-1223.

3.1.1.5 Towing devices. Not less than two hooks, loops, or pins for towing the vehicle shall be furnished on the front of the vehicle.

* 3.1.1.6 Wheel splash and stone throw protection. The vehicle shall have rigid splash shields ahead of the rear wheels and rubber mud flaps to the rear of the rear wheels. All tilt cabs shall have rubber mud flaps to the rear of the front wheels. A metal strip not less than 1/8 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1-1/4 inch wide by 3/32 inch thick shall be furnished at each bolt. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

* 3.1.1.7 Trailer towing package. A trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle, a trailer brake control system and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51335-1 for class C and D vehicles and to MS51118-1 for class E vehicles. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 1,800 pounds and a horizontal drawbar load of not less than 18,000 pounds for class C and D vehicles; and a vertical tongue load of not less than 4,000 pounds and a horizontal drawbar load of not less than 40,000 pounds for class E vehicles directly to the chassis rails. The rearmost portion of the pintle shall be forward, but not more than four inches forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The attachment devices for class C and D vehicles shall be capable of accommodating a standard grab hook (2-3/16 inches wide, 47/64-inch thick and 31/64-inch throat width) for a 3/8 inch chain; the

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attachment devices for a class E vehicle shall be capable of accommodating a standard grab hook (4-9/16 inches wide, 1-3/16 inches thick, 25/32-inch throat width) for a 5/8-inch chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle. The trailer brake control system shall conform to 3.4.11.3.

* 3.1.1.7.1 Pintle height. When specified (see 6.2), the pintle shall be mounted on an added member, braced to the main frame rails at a height above the ground level of 22 inches, plus or minus 2 inches, with the truck at curb weight. The mounting structure and welds shall be capable of transmitting a drawbar pull at least equal to the rated tongue and drawbar load of the pintle.

3.1.1.8 Trailer lighting cable. When specified (see 6.2), a trailer lighting cable conforming to SAE J1067 shall be furnished. The cable shall be coiled and shall be not less than 110 inches long when fully extended. Both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

* 3.1.1.9 Hydraulic tailgate. When specified (see 6.2), the vehicle shall be equipped with an electric motor driven hydraulic tailgate having not less than 2,000 pounds rated capacity. All hydraulic cylinders shall be provided with flow restrictors in the downport of the cylinders to prevent the tailgate from falling rapidly in the event of hydraulic system failure. The tailgate platform shall be of the ramping type and shall have a depth of not less than 32 inches exclusive of the ramp. The ramp shall taper down to the ground level to facilitate loading with wheeled hand carts. Platform loading area shall be of nonskid sheet steel. Latches to hold the tailgate during loading and unloading of the truck shall be provided. The tailgate shall have devices for holding the platform in stowed position for vehicle travel. When the tailgate is in loading position, the clearance between the rear edge of the vehicle and the tailgate shall be not more than 3/4 inch. Controls shall be mounted outside the body on the curbside of the vehicle. The vehicle ignition switch or a separate switch in the driver's compartment shall allow the driver to disconnect the power source to the tailgate. A plate describing operation of the hydraulic tailgate shall be provided and shall be in close proximity to the hydraulic tailgate controls. A decal conforming to type I or II, class 1 of MIL-M-43719 may be furnished in lieu of a plate.

3.1.1.9.1 Rear fold tailgate. Unless a fold-under type tailgate is specified (see 3.1.1.8.2), the tailgate shall fold vertically against the rear of the vehicle for travel. All tailgate operations shall be hydraulically powered or metered, providing for raising, lowering, folding and unfolding

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without manual assistance. The tailgate platform width shall be not less than 90 inches. Rear bumper and additional rear end protection need not be furnished.

3.1.1.9.2 Fold-under tailgate. When specified (see 6.2), the tailgate shall manually fold under the vehicle for travel and manually unfold for use. Hydraulically powered raising and metered lowering shall be provided. The tailgate platform width shall be not less than 72 inches. Rear bumperettes extending to the rear beyond the stowed tailgate shall be provided on each side of the rear, beyond the 72-inch platform width. Additional rear end protection need not be furnished.

3.1.1.10 Silicone brake fluid. When specified (see 6.2), and if available as manufacturer's standard or optional brake fluid, brake fluid conforming to MIL-B-46176 shall be provided in the hydraulic brake system. A tag shall be placed near the master cylinder stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176."

* 3.1.1.11 Hydraulic fluid identification plate. When a body hydraulic system is furnished and the hydraulic system requires fluid replenishment, a corrosion-resistant identification plate shall be furnished. The plate shall be installed near the filler cap and shall identify the type of hydraulic fluid to be used.

* 3.1.1.12 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise to any Federal Motor Carrier Safety Regulation referenced herein or to any other Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

* 3.1.1.13 4x4 conversion. The chassis manufacturer's standard 4x2 truck chassis may be modified to provide all wheel drive conforming to the requirements specified herein if:

(a) The conversion axle manufacturer's engineering department specifically approves and certifies that all such modifications meet the design requirements and standards of the conversion axle manufacturer. Certification shall be based on both design analysis and proving ground test reports, which shall be made available to the engineering and quality assurance offices of the procuring activity.

* (b) The chassis manufacturer's front axle before conversion has the same rating as the conversion axle to be installed.

* (c) Components used in the all wheel drive conversion are of current production.

* (d) Components used in the all wheel drive conversion are approved for the conversion application by the component manufacturers.

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* (e) The conversion vehicle is certified to conform to Federal Motor Vehicle Safety Standard, No. 121, by the intermediate or final manufacturer.
(f) Replacement headlights, if required to be added to meet the height requirements of Federal Motor Vehicle Safety Standard No. 108, are equivalent in mounting, protection, and range and precision of adjustment to the chassis manufacturer's original standard headlights.
(g) Unused headlight cavities are covered in a neat workmanlike manner, treated and painted to match the chassis cab color with treatment and painting equivalent to the chassis cab manufacturer's process for the remainder of the chassis cab. Cavities and their covers shall be rustproofed in accordance with 3.1.1.2.
* (h) Complete installation drawings are available to the procuring activity.

* 3.1.1.14 Decals and data plates. The hydraulic tailgate operating instructions and the power takeoff caution notice shall conform to type I or type II, class 1 of MIL-M-43719 or composition A (class l or 2) or composition C of MIL-P-5l4.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with the State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and the cab, with all attachments, accessories and equipment; the body; and a full complement of fuel, lubricants and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator weight (computed at 175 pounds), and a payload to provide not less than the specified GVW. Vehicle with four-door crew cab shall have GVW computed with 525 pounds of payload in the rear seat.

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3.2.6 Weight distribution. The distribution of the GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be made available to the engineering office of the procuring activity. Minimum GVW ratings shall conform to table II for the specified class of vehicle.

* TABLE II. GVW minimum ratings.

Vehicle class	GVW rating (pounds)
C	19,000
D	21,000
E	24,000

* 3.2.8 Overall width. The overall width of the vehicle, exclusive of tires and safety related items such as mirrors, lights and reflectors, shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

* 3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.10 Asbestos. Asbestos materials shall not be used in any form in any part of the vehicle.

3.3 Performance. The vehicle shall meet the following performance requirements. For 4x4 model vehicles, performance requirements shall be met with front wheel drive disengaged except that a vehicle equipped with interaxle compensating devices shall meet the performance with the front wheel drive engaged.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend continuous grades specified in table III at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive, and when multispeed axle is furnished, with the axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

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* TABLE III. Gradeability.

Vehicle class	Percent grade
C	1.5
D	1.5
E	1.0

* 3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with the engine operating at not less than 35 percent of recommended governed speed, and shall provide not more than the vehicle speed (mph) specified in table IV for the corresponding number of available forward speeds.

TABLE IV. Vehicle low speed requirement.

Model	Class	Number of forward speeds	Speed (mph)
4x2	C and D	4 or 5	4.5
		7 or 8	3.5
		9 or 10	3.0
	E	5	5.0
		7 or 8	4.0
		9 or 10	3.0
4x4	E	8	3.0
		9 or 10	2.5

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 58 mph for model 4x2 vehicles and not less than 55 mph for model 4x4 vehicles. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

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* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

* 3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at engine speed not more than manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operators manual. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level.

* 3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The clutch lining shall be asbestos free.

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3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.9 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater and a fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block including a fuse or circuit breaker to a single three-pronged (male), weatherproof slave receptacle for receiving external power and grounding vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

(a) A coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. Engine thermostat with an operating range of 170°F to 195°F shall be installed
(b) An immersion type engine oil heater, 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening
(c) The battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall include a thermostat to limit the temperature of the electrolyte to not more than 80°F
(d) A fuel warmer or pre-heater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a fuel flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical systems. Electrical systems shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

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* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with l2-volt dc lighting system, and not less than a 60-ampere alternator, shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable reservoir of not less than 12 fluid ounces and shall be furnished. A reservoir need not be furnished.

3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished. Except as otherwise specified, an alternator of not less than 40-ampere rated capacity and which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished. When an electric motor driven hydraulic tailgate is furnished (see 3.1.1.10), or when a converter type 24-volt system is provided for trailer connector (see 3.4.2.7), an alternator of not less than 60-ampere rated capacity, and which provides not less than 16 amperes dc output at normal engine idle speed, shall be furnished.

* 3.4.2.3 Lighting. All vehicle lights, reflectors, and wiring shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails or mounted on vehicle bumpers (unless recessed and fully protected).

* 3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The vehicle shall be provided with double-faced front signal units and single-faced rear signal units installed in accordance with SAE J588. Turn signal operating units shall be self-cancelling type with visible and audible type flash indicators.

3.4.2.5 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and total cold cranking ampere rating at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table V. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-l31.

TABLE V. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	320	1,200
Gasoline	100	450
* Gasoline (with electric tailgate)	115	520

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3.4.2.6 Auxiliary 24-volt system with trailer receptacle. When specified (see 6.2), an auxiliary 24-volt system with a trailer receptacle assembly shall be furnished. Either a converter type (see 3.4.2.7.1) or an alternator type (see 3.4.2.7.2) system meeting specified requirements shall be furnished. A trailer receptacle conforming to MS75021-2 with cover assembly shall be provided in an accessible location on the rear end of the vehicle. A twelve-conductor cable not less than 110 inches long, with both ends of the cable equipped with connectors conforming to MS75020-1 and MS75020-2 shall be furnished. The cable assembly shall be stowed in the vehicle. The 24-volt, 12-ampere output service lighting circuit shall be connected through the appropriate lighting controls to terminals B, D, E, J and L of MS75021-2.

3.4.2.6.1 Converter type 24-volt system. The 12- to 24-volt converter(s) shall operate from the 12-volt battery (see 3.4.2.6). The output capacity shall be not less than 12 amperes.

3.4.2.6.2 Alternator type 24-volt system. The alternator type 24-volt system shall be separate from the l2-volt vehicle lighting and ignition system and shall include:

(a) Nominal 24-volt alternator with not less than 25 amperes rated capacity and capable of providing not less than 7 amperes dc output at normal engine idle speed
(b) Two 12-volt batteries with a combined capacity of at least 40-ampere hours at a 20-hour rate or one 24-volt battery with 20-ampere hours capacity at a 20-hour rate
(c) Voltage regulating device
(d) An ammeter, mounted on the instrument panel.

3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tanks. Fuel tank(s) shall have a total capacity of not less than 50 gallons (43 gallons for vehicles destined for California). When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve, connecting either tank to the engine fuel intake, shall be provided and means shall be provided to monitor the level of either tank from a single fuel gage.

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3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. If vertical exhaust mufflers are furnished a heat shield shall be furnished. Vertical exhaust systems shall be provided with a hinged rain cap.

3.4.5 Gear train.

3.4.5.1 Manual transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided on the vehicle. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall provide for synchronized shift for at least the three highest forward speeds; or for class D and E vehicles a constant mesh transmission may be provided. Gear ratios in the transmission and axles shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide the vehicle performance specified in 3.3.1 through 3.3.1.3. The transmission shall be provided with power takeoff opening(s) conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds.

3.4.5.1.1 Speeds. The vehicle shall be provided with not less than the number of forward speeds specified in table VI.

TABLE VI. Minimum number of vehicle forward speeds.

Vehicle model	Vehicle class	Number of forward speeds
4x2	C	4
	D	5
	E	5
4x4	E	9

3.4.5.1.2 Transfer case. On 4x4 model vehicles, unless the transfer case is equipped with devices which compensate for differential torque and speeds between front and rear axles, the transfer case shall provide for driver selection of either two-wheel or four-wheel drive. When furnished, interaxle compensating devices shall provide for positive transfer of power to all driving axles. When a two-speed transfer case is furnished, the speedometer shall read accurate speed (mph) with the speed selector in high or low range.

* 3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the class and model vehicle and engine furnished, with the clutch torque capacity exceeding the maximum delivered engine torque. The clutch lining shall be of nonasbestos material.

3.4.5.3 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic or semiautomatic transmission. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the

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selector. The transmission shall be provided with an SAE J704 power takeoff opening. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

3.4.5.4 Power takeoff. When a power takeoff is furnished, it shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver (see 3.1.1.14).

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the class and model of the vehicle furnished. Chassis frame rails shall not project beyond the rear end of the body.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system, with components having a rated capacity at least equal to the load imposed on each member measured at the ground, with the vehicle loaded to specified GVW. When suspension is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. On nondriving axles, the hubcaps shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure build-up and for replenishing the oil supply shall be provided.

3.4.9.1 Two-speed axle. When specified for 4x2 model vehicles (see 6.2), a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. The gear ratio shall provide the performance specified in 3.3.1 through 3.3.1.3.

3.4.9.2 Traction control. The vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to ensure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

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* 3.4.10 Wheels, rims, tires, and tubes. Unless wide base tires are specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Rims and disc wheel sizes shall be the same for all wheels on each vehicle. Except when tube type tires are specified in procurement documents, multi-piece rims shall not be furnished. Tire size and load range (ply rating) shall be the same for all tires on the vehicle. When specified (see 6.2), wide base type tires and wheels for the front and rear axles shall be provided in lieu of conventional front and dual rear wheels and tires. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Steel belted radial ply, or when specified (see 6.2), bias ply tires shall be furnished. Unless otherwise specified, tires shall be tube or tubeless type with highway tread on 4x2 model vehicles, and with nondirectional mud and snow tread on 4x4 model vehicles. When specified (see 6.2), nondirectional mud and snow tires shall be furnished on the drive wheels and spare on 4x2 model vehicles. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to specified GVW. Tires and tire size designation systems shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

3.4.10.2 Inner tubes. When tube type tires are specified in procurement documents, inner tubes shall be of heavy duty type and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. The carrier design shall enable removal or mounting of a spare wheel assembly using only tools specified in 3.4.16.1. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material.

3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be installed on the carrier. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as furnished on the vehicle.

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3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

3.4.11.1 Air brakes. A four-wheel brake system consisting of full-air-application brakes shall be provided. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7-1/4 cubic feet per minute (cfm)
(b) Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
* (i) Automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes.

* 3.4.11.2.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.3 Trailer brake control system. In addition to the components specified in 3.4.11.2, a trailer brake control system shall be furnished. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Prime mover protection valve with dash control and automatic breakaway feature
(e) Trailer stoplight control operable with foot brake and with hand control for trailer brakes
(f) Two SAE J844 coiled air hoses, not less than 110 inches long when fully extended, with SAE J318 gladhand couplers on both ends of hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment

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(g) Air connectors for trailer with SAE J3l8 gladhand couplers mounted at the rear of the vehicle located to prevent interference with trailer
(h) Dummy gladhand couplers with chains
(i) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.4 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. When specified (see 6.2), a tilt cab with tilting and locking mechanism shall be furnished. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 339, including the first step height, which shall be not more than 24 inches. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with an external, key-operated lock. Drip rails shall be installed above the cab doors. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished. When specified for tilt cabs (see 6.2), vehicle with a height from the ground to the top of the windshield of more than 80 inches shall include provisions to facilitate cleaning of the windshield. Provisions shall consist of a bumper step or bumper step cutouts and a grab handle located under the windshield.

* 3.4.12.1 Cab interior. The cab shall have an upholstered, full-width, adjustable seat and back or individual, adjustable driver's seat and individual passenger seat. When specified (see 6.2), the driver's seat shall be the manufacturer's standard air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Two pairs of seat belts shall be installed.

* 3.4.12.2 Crew cab. When specified (see 6.2), a four-door, full width crew cab, shall be furnished in lieu of the standard full width cab. The cab shall be equipped with two upholstered full width seats and backs. The front seat shall be adjustable. Three pairs of seat belts shall be installed for both front and rear seats. The cab doors shall be equipped with locks operable from

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inside the cab through mechanical linkage, with both front doors equipped with external key-operated locks. Cab doors shall have windows with crank-operated window regulators. A rear window shall be provided. Interior lighting shall be provided. The cab roof shall be of one piece-construction or, if welded, the roof shall give the appearance of one piece, with weld seams being continuous, waterproof, and free of visible seams, bumps or protrusions. Full length drip moldings shall be mounted above the doors. When a crew cab is furnished, the cab-to-axle dimension shall be measured from the back face of the crew cab.

3.4.12.2.1 Crew cab seat spacing. With the front seat adjusted to the extreme forward position there shall be not less than 30 inches, measured in a horizontal plane, between the front of the rear seat-back and the rear of the front seat-back. A kick-space height of not less than 2-3/4 inches shall be maintained between the floor and the rear of the front seat in all positions of adjustment. Leg room and space forward of the front seat shall be equivalent to that provided ahead of the seat in a two-door standard cab.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be of the multispeed type and operated by either air or electric motor(s).

3.4.15 Bumper. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided.

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for tools may be furnished inside the cab. When stowage space for tools is located outside the cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with the tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

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3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left hand drive. The controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in the driving position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
* (d) Oil pressure gage or red indicator warning light
(e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Ash receptacle
(i) Dual sunvisors
(j) Front door mounted armrest on driver's side
(k) Driver's compartment ventilator other than window
* (l) Tachometer (diesel engine driven vehicle only).

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 20 square inches of reflective area. The mirrors shall have not less than two supports.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be furnished.

3.4.22 Engine hour meter. When specified (see 6.2), an engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel, or in the engine compartment in a readable location.

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3.4.23 Back-up alarm. The vehicle shall be provided with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

* 3.4.24 Front mounted winch. When specified (see 6.2), a winch assembly mounted on the front of the vehicle and powered by the manufacturer's standard power takeoff or powered hydraulically shall be provided. The winch shall conform to SAE J706. The winch shall be of the single drum type and shall have one forward and one reverse speed. All winch controls shall be mounted inside the cab. Winch controls shall be located to provide no interference with the entrance or exit of the driver. The winch shall have a rated single line pull capacity of not less than 20,000 pounds pull on the bare drum. The winch line speed on the bare drum shall be between 15 and 30 feet per minute at an engine speed equal to 35 percent of engine governed speed. The winch shall be wound with not less than 185 feet of 5/8-inch diameter, preformed 6x37, improved plow steel, independent wire rope core (IWRC) cable and shall be equipped with end chain and hook. An integral, adjustable, automatic safety brake shall be provided. The winch shall be equipped with a roller guide.

3.4.24.1 Winch drum guard. A winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges. The guard shall consist of not less than 1-1/4-inch vertical side plates, conforming to the outside radius of the drum flanges. Six bars, 3/8 inch by 1-1/4 inches, shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the drum flanges shall be not greater than one half the specified cable diameter.

3.4.24.2 Front bumper. When a front mounted winch is specified (see 3.4.24), the front bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be either channel (see 3.4.15) or a pipe type front bumper. When a pipe type front bumper is furnished, the nominal diameter shall be not less than 3 inches, and the wall thickness shall be not less than that specified in Schedule 40 of ASTM A120. The pipe type front bumper shall have half-round ball ends.

3.4.24.3 Combination step plate and gravel guard. When a front mounted winch is specified, a combination step plate and gravel guard shall be furnished. The step plate shall be fabricated of not less than 14 gage (0.0747 inch) steel tread plate exclusive of projections. The step plate shall be secured to the front bumper and shall be not less than 1/4 inch and not more than 3/8 inch from the chassis sheet metal. The step plate shall be capable of supporting 300 pounds per square foot. The step plate shall deflect not more than 1/8 inch under the loads imposed.

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3.5 Body and equipment. The vehicle shall have a cab-to-axle dimension of not less than 84 inches. When a crew cab is specified (see 3.4.12.2), the vehicle shall have a cab-to-axle dimension of not less than 83 inches, measured from the back face of the crew cab. The loadspace shall be not less than 147 inches in length and 88 inches in width. Body side panels shall be not less than 24 inches in height unless racks are specified (see 3.5.4).

* 3.5.1 Body construction. The body shall be of all-metal welded construction with members of minimum gage thickness as specified in table VII for carbon steel. High tensile steel may be furnished in two gages lighter weight, in accordance with U.S. Standard gage sizes.

* 3.5.2 Floor, sides and front. The floor shall be diamond tread plate. Drain holes shall be provided in the front of the body floor. The top edge of body side panels shall be flanged and shall have five openings provided on each side for body bows having cross-sectional dimensions of 1 inch by 2-1/2 inches. Lower bow sockets with bow stops shall be provided. Tiedown devices for securing a tarpaulin shall be provided. Structural sections of equal thickness may be used in lieu of framed sections specified in table VII. The front panel shall be capable of withstanding a horizontal static load equal to half the payload capacity of the vehicle without permanent distortion.

TABLE VII. Framing gages.

Framing member	U.S. Standard gage no.	Equivalent inches
Floor	8	0.1644
Tailgate	8	0.1644
Crossmembers	10	0.1345
Side and end rails	12	0.1046
Longitudinal sills	10	0.1345
Reinforcements	10	0.1345
Posts	12	0.1046
Side and front panels	12	0.1046

3.5.3 Tailgate. The tailgate shall be hinged at the bottom and provided with duck-covered stay-chains of a length to retain the tailgate in a horizontal position and in the vertical down position, at the operator's option. Steps shall be provided on each side of the tailgate to assist personnel in mounting and dismounting. The rear lights and reflectors shall be recessed in the rear bolster and shall be positioned to prevent damage by the lowering of the tailgate.

3.5.4 Body frame. The body shall be constructed with welded steel subframe. There shall be not less than 10 full-width crossmembers, including ends, and stub crossmembers as required to provide rigidity. The crossmembers shall be of full channel, pressed steel construction, reinforced by gusset plates or brackets at the point of attachment to the longitudinal sills. Crossmembers shall be welded to the body shell with not less than 4-inch lengths of weld, front and rear of both ends of each crossmember, and with staggered intermittent welds of not less

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than 4-inch lengths, on not more than l2-inch centers; or 2-inch lengths, on not more than 6-inch centers. Contact edges of the crossmembers with longitudinal sills and contact edges of welded reinforcements shall be welded for not less than 50 percent of the edge length.

3.5.5 body mounting. The body shall be secured with U-bolts, twin studs, brackets or J-bolts and shall include a wood breaker strip.

3.5.5.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than three U-bolts or twin studs per side on class A, B, C, and D vehicles and not less than four U-bolts or twin studs per side on class E vehicles, each having 9/16-inch minimum body diameter with 5/8-inch minimum thread diameter. The tie-plates shall be at least 1/2 inch thick. The vehicle chassis frame shall be braced using wood blocks at each mounting point unless mounting point is located at a full depth, frame crossmember. The blocks shall incorporate a keeper strap or groove for the mounting bolt, and shall be of a width and thickness to assure retention. Two tie-back straps shall be provided, one on each side of the rear portion of the body subframe, to maintain body alinement on the vehicle chassis. The forward body mounting bolts shall be located to the rear of the tapered portion of the breaker strips (see 3.5.3.3).

3.5.5.2 Brackets. When brackets are used, they shall be bolted to the web of the chassis frame rails. The body mounting brackets shall provide a means for drawing down the body on the chassis rails, and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure mounting brackets to chassis frame rails, they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builder's layout. Attachments shall neither interfere with nor obstruct chassis components.

3.5.5.3 J-bolts. When J-bolts are used, there shall be not less than seven J-bolts per side on class A, B, C and E vehicles, each not less than 3/8 inch diameter. J-bolts shall encircle the lower chassis frame flange. Two tie-back straps as specified in 3.5.3.1 shall be provided. The forward J-bolts shall be located to the rear of the tapered portion of the breaker strips (see 3.5.3.4). Provisions shall be made to prevent lateral shifting of the breaker strips.

* 3.5.5.4 Breaker strips. A hardwood or dense southern yellow pine breaker strip, of not less than 3/4 inch finished thickness, shall be installed between longitudinal sills and vehicle chassis frame. The minimum thickness specified shall be increased as may be required to provide adequate tire chain-to-body clearance. Breaker strips shall have a taper of not less than 1/2 inch in 16 to 18 inches at the forward end.

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* 3.5.6 Racks, troop seats, tarpaulin and bows. When specified (see 6.2), the vehicle shall be furnished with racks capable of folding down to form seats, a fitted tarpaulin, knockdown type bows, and tiedown devices. When racks are furnished, the height of the body side panels (see 3.5) shall be not less than 14 inches.

3.5.6.1 Racks. Removable racks, running the full length of the cargo body, shall be furnished on each side. The height of the racks shall be not less than 36 inches, measured from the floor. Steel upright posts shall be formed into box section pockets to take the top bows. Posts shall have provisions for drainage. Rack slats shall be of oak, not less than 3/4 inch to finished thickness. Width of individual slats shall be 3 inches. All slat edges shall be beveled or rounded. The racks shall have five or six slats with approximately 1 inch spacing between slats. The bottom three or four slats shall fold down to produce a seat. When in the folded-down position, the height of the seat shall be 17 inches, plus or minus 1 inch, measured from the floor at the front edge of the seat. The seat shall be level, or shall slope down towards the seat back, and shall be supported by folding braces. The seat shall have provisions for locking in the up position. Wood shall be treated (see 3.1.1.4).

* 3.5.6.2 Tarpaulin, bows and tiedown devices. The tarpaulin shall be fabricated of number 8 cotton duck conforming to type I of CCC-C-419. The tarpaulin material shall be water-repellant and fire-resistant. The tarpaulin shall be reinforced at the corners and other wearpoints with patches fabricated of the same base material as the tarpaulin. The tarpaulin color shall be olive drab. The rear curtain shall be of the roll-up type. The front curtain shall have a window not less than 12 by 24 inches in size, alined with the rear window of the vehicle cab. The tarpaulin shall completely cover the entire body and shall extend down the sides, front and rear, with the bows in place, to within three inches of the platform. Bows shall be on the outside of the racks and shall be constructed of metal or metal and hardwood components. At least five bows shall be furnished and shall provide an inside height of not less than 70 inches between the floor of the platform body and the tarpaulin cover of the top. Not less than five evenly spaced tiedown devices shall be provided on each side of the body.

3.5.7 Cargo tiedowns. When specified (see 6.2), cargo tiedown rings, fully recessed, with a 3-inch inside diameter and a 2,000-pound load rating, shall be installed in the floor of the loadspace as follows:

(a) One centered at the front of the loadspace
(b) Four at the outboard edge of each side, equally spaced the length of the loadspace.

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3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights, adjustment of the engine and brake systems; filling and charging of batteries; alinement of the front wheels; inflation of all tires; complete lubrication of chassis, engine and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

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4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on the front and rear axles. The total imposed loading on the front and rear axles shall be computed by the contractor and verified by the Government using the curb weight, the operator weight at 175 pounds, and the payload required to provide the specified GVW. Except as specified in 3.2.5, the calculated imposed loads on the front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

* 4.3.2 Road test. The vehicle, for models built to 4 wheel drive while on the chassis manufacturer's production line, shall be road tested by the contractor without payload. The road test shall be for not less than 10 miles at speeds up to 50 mph. The vehicle, for 4 wheel drive models qualifying under 3.1.1.13, shall be road tested with and without payload. Payload shall be distributed first, so that the front axle is loaded to GAWR and the vehicle is loaded to GVW and second, with the rear axle loaded to GAWR and the vehicle loaded to GVW. The road test for each of the three conditions shall be for not less than 30 miles at speeds up to 50 mph over highways and gravel roads and for not less than 5 miles at speeds up to 15 mph over cross country terrain with ground and grade requiring all wheel drive. During the loaded and empty portions of the road test, the brakes shall be applied firmly, bringing the vehicle to a sudden stop no less than 5 times during each portion of the road test. During the road test, the front and rear suspension and the drivetrain shall be periodically inspected for interference and contact with other vehicle components. Abnormal contact of the drivetrain or suspension components with other components shall be cause for rejection. Front spring bumpers shall not make contact with frame stops except under the most extreme cross country conditions. Operational or mechanical failures of vehicle components during the road test shall be cause for rejection. Failure includes permanent deformation as well as breakage.

4.3.3 Truck body equipment and painting. The certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to Government representatives at the first production vehicle inspection.

* 4.3.4 Wood treatment certification. Manufacturer’s records shall be available to verify that all wood requiring treatment in accordance with MIL-STD-1223 has been treated.

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4.3.5 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturer's facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government in transporting personnel and cargo.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and model of vehicle required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate service for painting and marking (see 3.1.1.1).
(e) Concealed markings, if required (see 3.1.1.1).
(f) Rustproofing, if required (see 3.1.1.2).
(g) Tropical rustproofing, if required (see 3.1.1.2).
* (h) Trailer towing package, if required (see 3.1.1.7).
(i) Pintle height of 22 inches, if required (see 3.1.1.7.1).
(j) Trailer lighting cable, if required (see 3.1.1.8).
(k) Hydraulic tailgate, if required (see 3.1.1.9).
(l) Fold-under type hydraulic tailgate, if required (see 3.1.1.9.2).

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(m) Silicone brake fluid, if required (see 3.1.1.10).
(n) Gasoline engine, if required (see 3.4.1.2).
(o) Silicone rubber radiator and heater hoses, if required (see 3.4.1.7).
(p) Power plant heaters and fuel warmer, if required (see 3.4.1.8).
(q) If a battery heater is not required with the power plant heaters (see 3.4.1.8).
(r) Auxiliary 24-volt system with trailer receptacle, if required (see 3.4.2.7).
(s) Automatic or semiautomatic transmission, if required (see 3.4.5.3).
(t) Oil lubricated wheel bearings and axle spindles, if required (see 3.4.9).
(u) Two-speed axle, if required for 4x2 model vehicles (see 3.4.9.1).
(v) Wide base type tires and wheels, if required (see 3.4.10).
(w) Disc type wheels, if required (see 3.4.10).
(x) Bias ply tires, if required (see 3.4.10.1).
* (y) Mud and snow tires on 4x2 vehicles, if required (see 3.4.10.1).
(z) A carrier for a spare tire assembly, if required (see 3.4.10.3).
(aa) Spare wheel or rim, if required (see 3.4.10.4).
(ab) A spare tire assembly, if required (see 3.4.10.5).
(ac) Brake controls for use from a towing vehicle, if required (see 3.4.11.4).
* (ad) Tilting hood and fender assembly, if required (see 3.4.12).
(ae) Tilt type cab, if required (see 3.4.12).
* (af) Air ride driver's seat, if required (see 3.4.12.1).
(ag) Tilt type cab windshield cleaning step and grab handle, if required (see 3.4.12).
(ah) Crew cab, if required (see 3.4.12.2).
(ai) Tools, if required, (see 3.4.16.1).
(aj) Air operated horn, in addition, if required (see 3.4.21).
(ak) Engine hour meter, if required (see 3.4.22).
(al) Front mounted winch, if required (see 3.4.24).
(am) Racks, troop seats, tarpaulin and bows, if required (see 3.5.6).
* (an) Cargo tiedown rings, if required (see 3.5.7).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number and vehicle type and class. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.

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Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.00
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, tabulated 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speeds. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0

* 6.4 Subject term (key word) listing.

Tarpaulin
Troop seats
Troop transport
Truck, cargo
Truck, commercial
Truck, nontactical
Winch.

6.5 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

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